

EDUCATION	Arizona State University	Tempe, AZ
	<i>Ph.D. Student in Computer Science</i>	2023 - present
	• Research area: Software Engineering, Program Analysis, Machine Learning • <i>Advisor: Prof. Xusheng Xiao</i>	
	University of Washington	Seattle, WA
	<i>Master of Science in Electrical & Computer Engineering</i>	2019 - 2021
	Cuiying Honors College, Lanzhou University	Lanzhou, China
	<i>Bachelor of Science in Physics, with highest Honors</i>	2015 - 2019
PUBLICATIONS	1. Shang Ma, Chen Yanai, Avichai Ben, Zichen Liu , Yanfang Ye, Xusheng Xiao. “Demystifying Scam Playbooks: Psychological Techniques, Scenario Taxonomy, and Conversation-Aware Detection.” <i>Under submission to the IEEE Symposium on Security and Privacy (S&P)</i> , 2027. [arXiv]	
	2. Zichen Liu and Xusheng Xiao. “AppBDS: LLM-Powered Description Synthesis for Sensitive Behaviors in Mobile Apps.” <i>Proceedings of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE)</i> , 2025. [Paper]	
	3. Zichen Liu , Shao Yang, Xusheng Xiao. “BINCTX: Multi-Modal Representation Learning for Robust Android App Behavior Detection.” <i>arXiv preprint arXiv:2510.14344</i> , 2025. [arXiv]	
	4. Jiong Xiao Wang*, Zichen Liu* , Keun Hee Park, Muhao Chen, and Chaowei Xiao. “Adversarial Demonstration Attacks on Large Language Models.” <i>arXiv preprint arXiv:2305.14950</i> , 2023. [arXiv]	
	5. Cheng-Yen Yang*, Aotian Zheng*, Zichen Liu , and Jenq-Neng Hwang. “Tracklet-based Multi-camera Multiple People Tracking.” In <i>International Conference on Computer Vision (ICCV) Workshops</i> , 2021. Ranked 3rd out of ~80 competitors. [Presentation]	
RESEARCH EXPERIENCE	Research Assistant RISE Lab @ASU	
	<i>Advisor: Prof. Xusheng Xiao</i>	2023.7 - present
	<i>Demystifying Scam Playbooks: Scenario Taxonomy and Conversation-Aware Detection</i>	
	Research Summary: Scammers embed reusable psychological techniques within recurring operational scenarios. This work characterizes scam scenarios at scale from real-world incident reports and turns that scenario-level knowledge into defenses for financial-institution customer interactions. • Designed and built EARS, a conversation-aware detection system that infers the underlying scam scenario mid-conversation, transferring scenario knowledge distilled from 102K real-world incident reports to financial-institution customer-service interactions • Introduced phase-aware turn modeling and a label-preserving LLM augmentation pipeline that let the model withhold judgment on benign turns and commit once diagnostic evidence appears, lifting scenario macro-F1 from 64.9 to 71.1; 84.3% tactic accuracy on 1,115 industry-partner conversations (14K turns), over 2× the macro-F1 of zero-shot LLM baselines • [<i>Ongoing</i>] Leading a longitudinal measurement study of the scam ecosystem over a self-built 611K-report, 2018–2026 corpus, tracing how scenarios, campaign organization, contact channels and persuasion effectiveness co-evolve, under a pre-registered and adversarially audited analysis protocol	

Fix Patterns and Security Impact for LLM-Guided Repair [Ongoing]

- Building a two-axis knowledge base over 9.8K CVEs from PatchDB, CVEfixes, MegaVul and ARVO: a mechanism-level security-impact taxonomy specifying the safety constraint each vulnerability violates, paired with a fix-pattern taxonomy mined from diff-AST representations of real patches
- Built the end-to-end evaluation chain from patch representation to generated-patch judging (7.6K patches, bug-level leakage splits) and showed that cluster purity—the metric this line of work had been optimizing—does not predict repair success, while cluster coverage does

AppBDS: LLM-Powered Description Synthesis for Sensitive Behaviors in Mobile Apps

Research Summary: Mobile apps increasingly access sensitive user data through dangerous permissions, yet only 37% adequately explain how that data is used. AppBDS closes this transparency gap by combining program analysis with LLMs to automatically describe sensitive permission usage.

- Built AppBDS, a framework pairing program analysis with a multi-agent LLM architecture for cross-validated, evidence-based description synthesis, backed by a 3,500-app knowledge base across 7 permission types, and outperforming state-of-the-art approaches in richness, specificity, and semantic relatedness
- Designed a UI-Fused Call Graph technique fusing static analysis with Android UI context, filtering 46,282 nodes down to 43 privacy-relevant ones; validated on 270 real-world apps across GPT-4o, LLaMA, and Qwen, and under code obfuscation

Research Assistant | SaFo Lab @ASU

Advisor: Prof. Chaowei Xiao

2023.2 - 2023.7

Adversarial Demonstration Attacks on Large Language Models

- Pioneered a novel threat model against LLM in-context learning, showing that manipulating demonstrations alone induces targeted misclassification with high success on GPT2-XL and LLaMA across multiple tasks
- Established the stealthiness of the attack via linguistic/semantic analysis, model perplexity, and user study

Research Assistant | Information Processing Lab @UW

Advisor: Prof. Jenq-Neng Hwang

2020.12 - 2021.9

Multi-camera Multiple People Tracking and Human Pose Estimation

- Built a multi-camera tracking system pairing a FairMOT baseline with a novel cross-camera tracklet association algorithm, placing **3rd (Track 2)** and **5th (Track 1)** in the AICITY Challenge at the ICCV 2021 Workshop against 70+ teams
- Proposed **Joint-Flow**, a tri-branch architecture fusing top-down, bottom-up, and temporal flow cues for robust pose tracking (**78.7 mAP**, **63.6 MOTA** on PoseTrack)

Machine Learning Engineer | Ernst Leitz Labs, Leica Camera AG 2021.09 - 2022.11

- Investigate and develop deep learning based method to replace and optimize mobile camera Image Signal Processor (ISP) pipeline.
- Participate in Leitz Looks Engine project: develop image rendering engine and depth estimation algorithms.
- Integrate various depth estimation algorithms such as RAFT-Stereo and MiDaS onto next generation Qualcomm Snapdragon chipset, and perform speed/power optimizing and performance profiling.

INDUSTRY
EXPERIENCE

TEACHING
EXPERIENCE

Teaching Assistant | Arizona State University

- CSE 464 Software Quality Assurance and Testing Fall 2023 – Fall 2025
- CSE 471 Introduction to Artificial Intelligence, Spring 24

ACADEMIC
SERVICE

- **Artifact Evaluation Committee:** IEEE Symposium on Security and Privacy (S&P) 2027, ISSTA 2026, ASE 2026
- **Reviewer:** NeurIPS 2026 FLLMPT Workshop

SKILLS	Languages: Python, Java, C/C++	
	AI & LLM: PyTorch, Hugging Face, Fine-tuning (LoRA/PEFT), vLLM, LangChain, RAG, Multi-Agent LLM Systems	
	Program Analysis & Security: Soot/SootUp, FlowDroid, Android/APK Analysis, Call-Graph & Taint Analysis	
	Tools & DevOps: Git, Docker, CI/CD	
PROJECTS		
	<i>Blockchain Consensus Simulator</i>	2020.10 - 2020.12
	Developed a Python simulator to benchmark consensus protocols (e.g., Conflux, Prism), validating their theoretical performance on throughput and latency.	
	<i>Adversarial Attacks and Defenses: A Survey</i>	2020.5 - 2020.6
	Authored a comprehensive survey on adversarial ML, synthesizing and categorizing key attack/defense methodologies and identifying research gaps.	